

A hybrid stochastic car-following model based on data-driven and theory-driven method

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Abstract—As a fundamental framework in micro traffic flow analysis, car-following (CF) behavior modeling is essential for understanding the mechanisms underlying traffic congestion formation and evolution. Data-driven CF models, based on machine learning, offer high prediction accuracy but lack interpretability and safety. In contrast, theory-driven CF models, rooted in micro traffic flow theory, provide clear physical interpretability and inherent safety but exhibit limited predictive accuracy. Furthermore, deterministic CF models struggle to capture stochastic factors influencing human driving behavior. To overcome these limitations, this study proposes a 2I-PBL hybrid stochastic CF model, integrating the data-driven PSO-Bi-LSTM model with the stochastic theory-driven 2D-IDM model. Firstly, the Bi-LSTM model is enhanced using the particle swarm optimization (PSO) algorithm, resulting in the PSO-Bi-LSTM machine learning CF model with superior predictive performance. Secondly, a combined forecasting method is employed to fuse the PSO-Bi-LSTM and 2D-IDM models, incorporating stochastic factors. This integration produces the 2I-PBL hybrid model, which accurately predicts driving behavior while ensuring safety. Finally, the proposed 2I-PBL model is validated by demonstrating its effectiveness in capturing heterogeneous driving behaviors.

Index Terms—car-following, Bi-LSTM, 2D-IDM model, PSO algorithm

I. INTRODUCTION

Car-following (CF) behavior is a fundamental element in microscopic traffic flow modeling, playing a critical role in congestion analysis, traffic management, and autonomous vehicle control. CF models are generally classified into two categories: theory-driven and data-driven. Theory-driven models, such as the early works of Reuschel [1], Pipes [2], and the Optimal Velocity (OV) model [3], simulate driver behavior based on vehicle dynamics and psychological assumptions. The Intelligent Driver Model (IDM) [4] and its variants, such as the Full Velocity Difference (FVD) model [5], have further improved behavioral realism by incorporating speed, headway, and velocity differences. Extensions like stochastic IDM and 2D-IDM [6] address the limitations of deterministic models by introducing behavioral randomness, enabling better representation of traffic oscillations and heterogeneous driving styles.

Parallel to these developments, data-driven models have gained popularity due to their ability to extract complex nonlinear patterns from empirical data. Traditional approaches using k -nearest neighbors, support vector regression (SVR) [7],

and shallow neural networks have evolved into more advanced deep learning architectures, such as RNNs [8], GRUs [9], and LSTMs [10]. These models demonstrate strong predictive performance but often lack interpretability and fail to incorporate physical driving constraints.

To leverage the strengths of both paradigms, hybrid CF models have emerged. These include combination forecasting [11], adaptive weighted integration [12], and physics-informed deep learning (PIDL), which embed physical knowledge into learning frameworks. For instance, the jointly-trained physics-guided LSTM (JTPG-LSTM) [13] balances accuracy and safety by coupling learning-based models with theory constraints. However, many existing hybrid models still adopt deterministic theory components, which limits their ability to simulate the inherent stochasticity and heterogeneity of real-world driving.

To address these limitations, this study proposes a hybrid stochastic CF model named 2I-PBL, which integrates a stochastic theory-driven model (2D-IDM) with a Bi-LSTM network optimized by particle swarm optimization (PSO). The 2D-IDM component ensures interpretability and captures behavioral randomness, while PSO enhances the Bi-LSTM's predictive accuracy. Experimental validation on the NGSIM dataset demonstrates that the 2I-PBL model outperforms traditional models and existing hybrids in capturing asymmetric and diverse driving behaviors.

The main contributions of this paper are as follows: (1) development of a GA-optimized stochastic 2D-IDM model with improved realism; (2) proposal of a PSO-enhanced Bi-LSTM framework embedded in a hybrid 2I-PBL architecture; and (3) empirical validation of the model's superior performance in simulating heterogeneous car-following dynamics under real-world traffic conditions.

II. METHODOLOGY

A. Car-following model based on theory-driven

Considering the importance of random factors in modeling actual driving behaviors, we selected 2D-IDM model. The model introduces a random expected time headway based on the IDM model, enabling precise simulation precisely simulate concave growth pattern of traffic oscillation and actual traffic

phenomena. The mathematical expressions for IDM model is as follows.

$$a_i(t) = a \left(1 - \left(\frac{v_i(t)}{v_{\max}} \right)^4 - \left(\frac{d_{i,desired}(t)}{d_i(t)} \right)^2 \right), \quad (1)$$

$$d_{i,desired}(t) = s_0 + v_i(t)T + \frac{v_i(t)\Delta v_{i,i-1}(t)}{2\sqrt{ab}}.$$

Among them, i represents the vehicle number, a is the maximum acceleration, b is the safety deceleration, $V_{\max}$ is the maximum speed, $d_i(t) = x_{i-1}(t) - x_i(t) - l$ is the space headway between the front car and the following car, $\Delta v_{i,i-1}(t) = v_i(t) - v_{i-1}(t)$, l is the length of the vehicle, s_0 represents the minimum space headway, T is the expected time headway.

In traditional IDM model, the expected time headway is constant, while the perception and response of different drivers to the time headway are different in the actual driving environment. Based on this, the 2D-IDM allows the expected time headway T varies from T_1 to T_2 in each simulation time step Δt with a probability of p .

$$T(t + \Delta t) = \begin{cases} T_1 + r(T_2 - T_1) & \text{IF } r_1 < p, \\ T(t) & \text{otherwise,} \end{cases} \quad (2)$$

where r and r_1 are two independents uniformly distributed random numbers between 0 and 1.

B. Car-following model based on machine learning

Bi-LSTM network is an advanced neural network structure built on the foundation of the above traditional LSTM. Unlike the unidirectional LSTM that can only process time series data from one direction, the Bi-LSTM model processes input data simultaneously in both directions. This dual-direction data processing method enables the Bi-LSTM network to comprehensively consider the information from both ends of the sequence when updating the content of the cell unit, thereby capturing more abundant and comprehensive sequence features and improving forecast precision. Bi-LSTM network is composed of two unidirectional LSTM layers. With their information flow directions opposite to each other. LSTM layers in these two directions are independent and do not share parameters such as weights and biases. In other words, the Bi-LSTM network actually contains two distinct sets of training parameters. One is responsible for controlling how the data information propagates forward, and the other is responsible for controlling how the data information propagates backward. This design allows the Bi-LSTM network to simultaneously learn forward and backward dependencies, enhancing its understanding of the sequence data by integrating information in both directions. At time t , Bi-LSTM model abstracts information characteristics from the front to the back and from the back to the front, respectively, to obtain forward and backward hidden state h_t^f and h_t^b . These are then concatenated to form $[h_t^f, h_t^b]$, which encapsulates all forward and backward information.

C. PSO algorithm

The performance of Bi-LSTM model is directly influenced by the number of iterations and the neurons in hidden layer. Proper hyper-parameter tuning is essential to achieve optimal model performance, these hyper-parameters generally require to be manually set, which may lead to suboptimal prediction outcomes of the model. To address this issue, we employ the PSO algorithm to optimize the hyper-parameters of Bi-LSTM network to enhance forecast accuracy. PSO algorithm is meta-heuristic optimization algorithm, inspired by the social behavior of birds or fish. The optimization method begins with the random solution and iteratively searches for the optimal solution. It is characterized by fast convergence, minimal parameter requirements, and simplicity. The particle update formula of the standard PSO algorithm can be expressed as follows.

$$\begin{aligned} v_i^{t+1} &= \omega v_i^t + c_1 r_1 (p_{best}^t - x_i^t) + c_2 r_2 (g_{best}^t - x_i^t) \\ \chi_i^{t+1} &= \chi_i^t + v_i^{t+1} \end{aligned} \quad (3)$$

In Equation 3, t represents the number of iterations, ω denotes the inertia weight, r_1 and r_2 are random numbers between (0, 1), and c_1 and c_2 represent self-awareness efficiency and social-awareness efficiency. p_{best} represents the particle's personal best position, while g_{best} represents the global best position of the swarm. To enhance global search capability and convergence velocity of standard PSO algorithm, we employ linear decreasing weight method, and express the weight ω as the form of Equation 4. The introduction of linear decreasing weights allows the algorithm gradually reduce the velocity of particles in the iterative process, enabling a more refined search. Simultaneously, this approach helps prevent the algorithm from falling into local optima, making it well-suited for solving complex high-dimensional and nonlinear optimization problems.

$$\omega = \omega_{\max} - \frac{t(\omega_{\max} - \omega_{\min})}{t_{\max}} \quad (4)$$

where $\omega_{\max}$ and $\omega_{\min}$ denotes maximum and minimum values of ω , $t_{\max}$ represents maximum number of iterations.

D. The establishment of the hybrid car-following model

Different prediction methods capture distinct information characteristics, and a single prediction model often fails to fully utilize all relevant data. To leverage the strengths of various models to improve forecast accuracy and information utilization efficiency, a combined prediction method was proposed. In order to accurately simulate the drivers' actual driving behavior, in this study, the combination forecasting method is employed to construct a hybrid car-following model. This model fully integrates the advantages of machine learning model and theory-driven model, and ensuring controllability and safety while accurately simulating the actual driving behavior. First, it is assumed that M model is used to predict sample y_t , and predictive rate of j_{th} model for y_t is y_{jt} . Forecast error of j_{th} model can be expressed as:

$e_{jt} = y_t - y_{jt}$. where, $t = 1, 2, \dots, N$, N is number of samples, $j = 1, 2, \dots, M$. Let $\omega_1, \omega_2, \dots, \omega_M$ be the weight coefficients corresponding to the M models in hybrid model. In order to guarantee unbiasedness of the hybrid model, weight coefficients should satisfy the following formula: $\omega_1 + \omega_2 + \dots + \omega_M = 1$, Predictive rate $\hat{y}_t$ of hybrid model for sample y_t can be expressed as follows:

$$\hat{y}_t = \omega_1 y_{1t} + \omega_2 y_{2t} + \dots + \omega_M y_{Mt} = \sum_j^M \omega_j y_{jt} \quad (5)$$

Predictive error of hybrid model for sample y_t could be calculated according following formula: $e_t = y_t - \hat{y}_t$. Then total error E of all objects could be calculated as follows: $E = \sum_{t=1}^N e_t^2$.

The problem of determining the optimal weights $\omega_1, \omega_2, \dots, \omega_M$ of each model in the hybrid model could be transformed into an optimization problem of solving minimum value of total error E . Output of hybrid model constructed by above approach represents the best combination of theory-driven car-following model and machine learning car-following model. Moreover, most models only focus on accuracy of speed forecast, and neglecting the importance of forecast accuracy of space headway, which may lead to deficiencies in the safety and controllability of the model. Therefore, to enhance the hybrid model's properties, we consider both speed and space headway in the total error E , leading to the following objective function:

$$\begin{aligned} \min E &= \beta E_v + (1 - \beta) E_g \\ &= \beta \sum_{t=1}^N e_{t-v}^2 + (1 - \beta) \sum_{t=1}^N e_{t-g}^2 \\ &= \sum_{t=1}^N (\beta e_{t-v}^2 + (1 - \beta) e_{t-g}^2) \\ &= \sum_{t=1}^N [\beta (y_{t-v} - \hat{y}_{t-v})^2 + (1 - \beta) (y_{t-g} - \hat{y}_{t-g})^2] \\ &= \sum_{t=1}^N [\beta (y_{t-v} - \sum_j^M \omega_j y_{jt-v})^2 + (1 - \beta) (y_{t-g} - \sum_j^M \omega_j y_{jt-g})^2], \\ \text{s.t. } \sum_{j=1}^M \omega_j &= 1; \omega_j \geq 0; 0 \leq \beta \leq 1. \end{aligned} \quad (6)$$

In the formula, E_v is the total error between the model predicted speed and the actual speed, E_g is the total error between the model predicted space headway and the actual space headway, β and $1 - \beta$ are the weights of the two measurements of error, y_{t-v} and y_{t-g} are the actual values of the t_{th} speed sample and the t_{th} space headway sample, respectively. Output of hybrid model can be regulated through modifying values of β and ω_j , where the value of β is adjusted manually according to the specific requirements, and the value of ω_j is obtained by solving Equation 6.

III. RESULT ANALYSIS

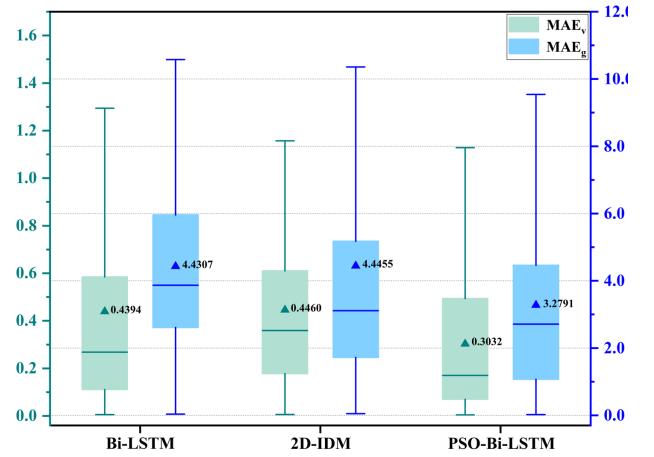
A. Model evaluation results

Considering that the genetic algorithm can efficiently obtain the global optimal solution and solve complex optimization problems [49], this study employs this algorithm to calibrate the parameters of the 2D-IDM model. In the 2D-IDM model, T_1 is set to 0.5 s, T_2 is set to 1.9 s, and p is set to 0.015. The parameter calibration results of the 2D-IDM model in this study are presented in Table I.

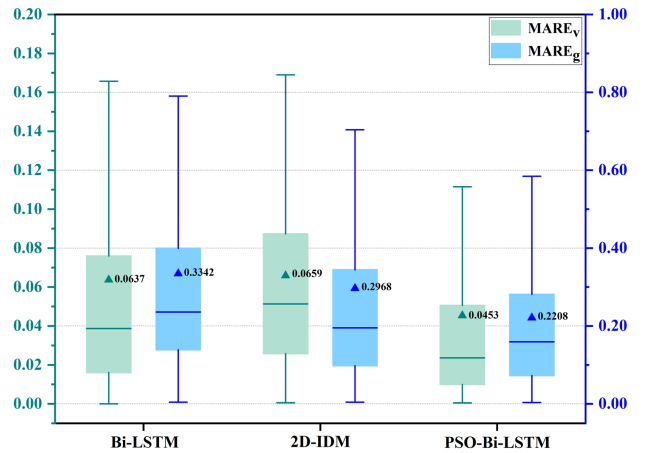
TABLE I: The calibration results of 2D-IDM model

Parameters	V_f	S_0	A_n	b_n
Value	0.317	2.74	22.17	1.96

We conduct a detailed comparison of the performance of Bi-LSTM, 2D-IDM and PSO-Bi-LSTM models in detail. The evaluation outcomes are illustrated in Figure 1 and the performance indices of the models exhibit different mean distributions.



(a) MAE distributions of models.



(b) MARE distributions of models.

Fig. 1: Distributions of evaluation indicators for each model (Triangles represent mean values).

To further explore forecasting ability of each model for car-following behavior, this section compares the error distribution intervals of evaluation indexes MAE_v , MAE_g , $MARE_v$, and $MARE_g$ of Bi-LSTM, 2D-IDM, and PSO-Bi-LSTM models in detail, as illustrated in Table II. The results show that the PSO-Bi-LSTM model demonstrates significant advantages in prediction accuracy and stability of prediction results, which proves the effectiveness of machine learning methods in simulating driving behavior. Table III analyzes the performance of the 2I-PBL hybrid model under various β values. We could find that when value of β changes within the range of 0 to 1, the weight ω_2 of PSO-Bi-LSTM model always exceeds 0.5. This finding confirms that the proposed machine learning model has better forecast ability than the 2D-IDM model in simulating the driver's actual driving behavior. By adjusting value of parameter β , the output of the 2I-PBL model can be flexibly controlled, enabling the establishment of a safe and controllable hybrid model with high forecasting precision.

TABLE II: Statistics results of Bi-LSTM, 2D-IDM, and PSO-Bi-LSTM(PBL) models.

	Models	Mean	Min	Max	Percentile [25%, 50%, 75%]
MAE_v	Bi-LSTM	0.4394	0.0056	10.5220	[0.1113, 0.2682, 0.5845]
	2D-IDM	0.4460	0.0062	11.8362	[0.1781, 0.3591, 0.6097]
	PBL	0.3032	0.0044	10.4200	[0.0702, 0.1700, 0.4935]
MAE_g	Bi-LSTM	4.4307	0.0371	25.6867	[2.6130, 3.8663, 5.9585]
	2D-IDM	4.4455	0.0482	24.6486	[1.7233, 3.1099, 5.1765]
	PBL	3.2791	0.0237	18.6093	[1.0733, 2.7133, 4.4611]
$MARE_v$	Bi-LSTM	0.0637	0.0006	4.4387	[0.0161, 0.0387, 0.0759]
	2D-IDM	0.0659	0.0006	4.4513	[0.0258, 0.0513, 0.0873]
	PBL	0.0453	0.0005	4.3798	[0.0100, 0.0236, 0.0506]
$MARE_g$	Bi-LSTM	0.3342	0.0043	4.5745	[0.1390, 0.2358, 0.3995]
	2D-IDM	0.2968	0.0042	4.2445	[0.0980, 0.1951, 0.3445]
	PBL	0.2208	0.0036	2.2159	[0.0726, 0.1593, 0.2814]

TABLE III: Evaluation results of 2I-PBL model.

β	2D-IDM	PSO-Bi-LSTM	Velocity		Gap	
	ω_1	ω_2	MAE_v	$MARE_v$	MAE_g	$MARE_g$
0	0.497	0.503	0.28726	0.04268	2.87529	0.18485
0.1	0.495	0.505	0.28688	0.04262	2.83263	0.18513
0.2	0.490	0.510	0.28617	0.04253	2.86561	0.18602
0.3	0.482	0.518	0.28459	0.04242	2.85528	0.18625
0.4	0.470	0.530	0.26335	0.04219	2.84046	0.18667
0.5	0.456	0.544	0.27988	0.04202	2.82609	0.18700
0.6	0.440	0.560	0.27646	0.04181	2.80631	0.18761
0.7	0.424	0.576	0.27293	0.04149	2.78828	0.18786
0.8	0.406	0.594	0.26986	0.04103	2.77616	0.18804
0.9	0.377	0.623	0.26665	0.04003	2.77440	0.18880
1	0.288	0.712	0.26565	0.03942	2.80750	0.18990

B. Optimal combination result

This paper uses sum of index $MARE_v$ and $MARE_g$ as measure of overall error of 2I-PBL hybrid model. The adjustment range of weight β is set to 0 to 1, and Figure 2 illustrates the variation in model's total error. According to the results, when the weight β is set to 0.864, hybrid model's total error achieves the smallest total error. Under this condition, the weights ω_1 and ω_2 of 2D-IDM model and the PSO-Bi-LSTM model are 0.403 and 0.597. Hybrid model under this weight combination can effectively reflect the driver's car-following behavior while ensuring both precision and efficiency.

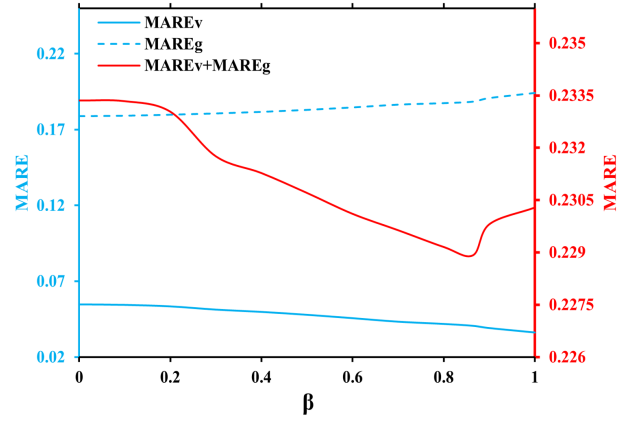
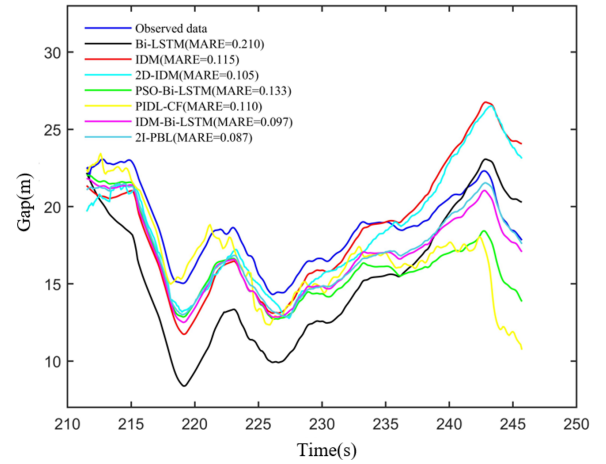
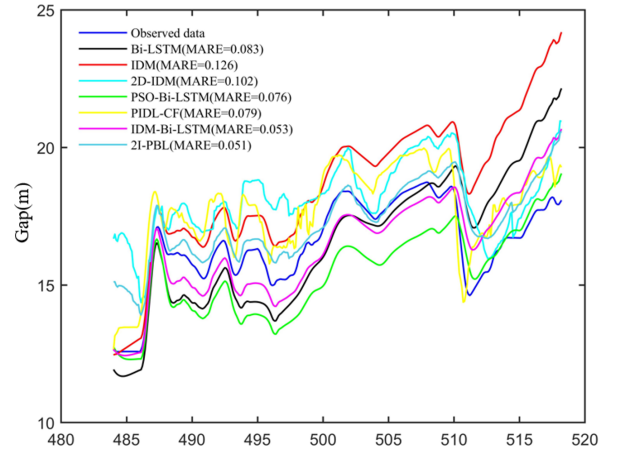


Fig. 2: Optimal combination result

To comprehensively evaluate validity of 2I-PBL hybrid model, in this part, we select the space headway data of two vehicle trajectory samples (vehicles 930 and 1284) to test model's performance to simulate car-following behavior. The simulation results are illustrated in Figure 3.



(a) Vehicle 930.



(b) Vehicle 1284.

Fig. 3: The simulated and actual gap curves.

The results indicate that when predicting space headway, 2D-IDM and 2I-PBL hybrid models exhibit smaller prediction errors compared to Bi-LSTM, PSO-Bi-LSTM, and IDM-Bi-LSTM models. In the first half of trajectory curve, prediction results of the 2D-IDM model are quite close to the actual data, but in the second half, significant deviations are observed. In contrast, the hybrid model achieves higher prediction accuracy and stability throughout the whole trajectory, and the error is also the smallest of all models. In Figure 3b, the simulated trajectory of hybrid model aligns more closely with observed data, demonstrating superior performance than other models, and also has the smallest error. These findings confirm that the forecast precision of 2I-PBL hybrid model is significantly higher than Bi-LSTM, 2D-IDM, PSO-Bi-LSTM PIDL-CF and IDM-Bi-LSTM models when modelling real trajectory data.

C. Analysis of heterogeneous driving behaviors

The heterogeneity of driving behavior is reflected in the differing strategies adopted by drivers under identical conditions. This study examined the capability of various models to capture such behavioral differences, categorizing car-following behavior into three types: normal, aggressive, and timid. These categories represent distinct driver preferences—ranging from maintaining moderate to very short or large space headways.

Simulation results (as shown in Figure 4) demonstrate that the proposed hybrid model outperforms traditional and baseline models in predicting position under all three behavior types. In particular, the PSO-Bi-LSTM model shows significant improvement over the standard Bi-LSTM, validating the effectiveness of the PSO optimization. Furthermore, the 2I-PBL model achieves the lowest prediction error among all models, confirming its superior capability in modeling heterogeneous driving behaviors.

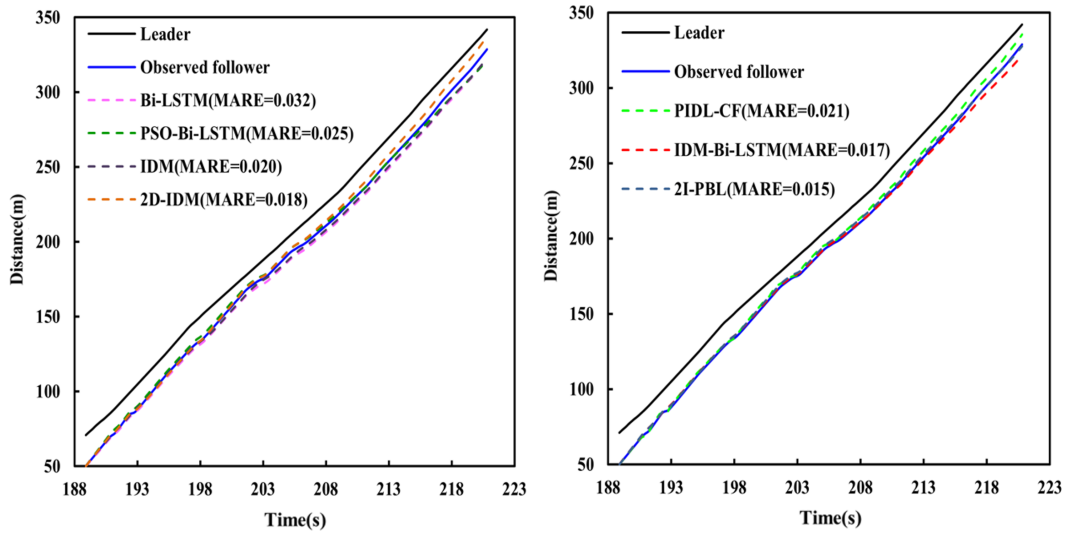
IV. CONCLUSIONS

To enhance the accuracy of car-following behavior modeling, this study proposes a hybrid approach that combines machine learning with theory-driven modeling. A particle swarm optimization (PSO) algorithm is employed to improve the performance of a Bi-LSTM neural network, which is subsequently integrated with the stochastic 2D-IDM model to develop the 2I-PBL hybrid car-following model. The model is trained and tested on the NGSIM dataset and evaluated for its effectiveness in capturing heterogeneous driving behaviors.

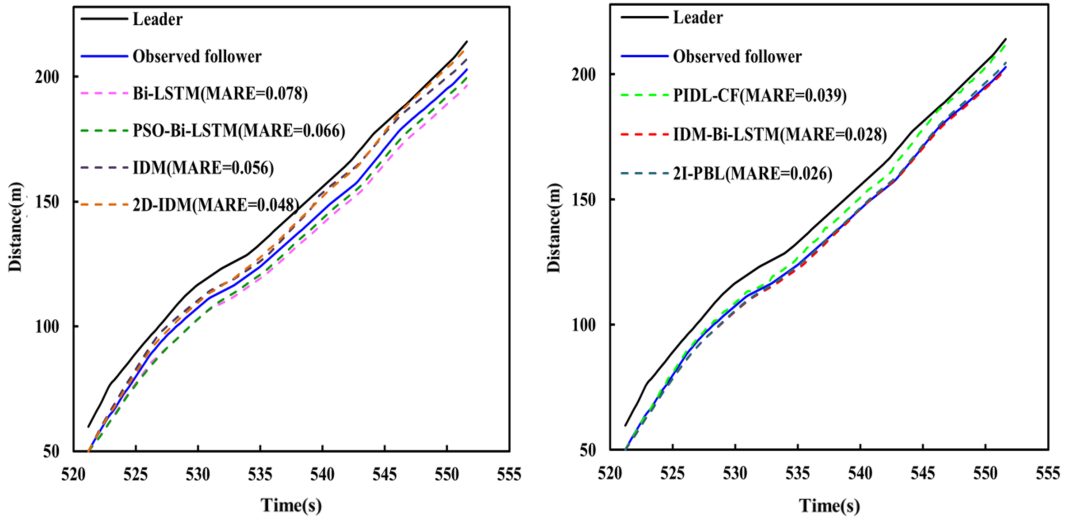
The results demonstrate that the PSO-Bi-LSTM model significantly improves upon the baseline Bi-LSTM, validating the effectiveness of PSO-based optimization. By introducing a weight parameter β , the hybrid model flexibly balances velocity and headway prediction. Comparative evaluations show that the 2I-PBL model consistently outperforms both traditional theory-driven models and recent hybrid approaches, offering enhanced fidelity in simulating real-world driving behavior. In addition, the model successfully replicates observed hysteresis loops, confirming its robustness under varying traffic conditions.

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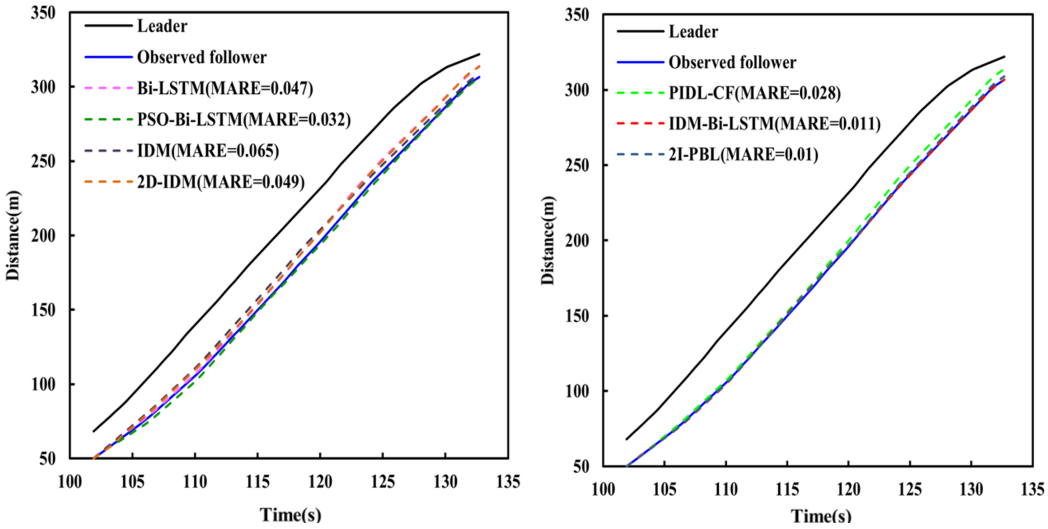
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(a) Normal driving behavior: Vehicle 873 (average gap 15.01 m).



(b) Aggressive driving behavior: Vehicle 1414 (average gap 9.50 m).



(c) Timid driving behavior: Vehicle 619 (average gap 31.36 m).

Fig. 4: Model's performance to simulate heterogeneous driving behaviors.